GraphRAG: Retrieval-Augmented Generation with Graph-Based Evidence Aggregation

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Abstract

We present GraphRAG, a novel method combining dense retrieval with graph-based evidence aggregation for multi-hop question answering. Experiments on three benchmarks show 12.3% accuracy improvement over dense retrieval baselines. Our approach is particularly effective for cross-document reasoning tasks.

1. Introduction

Retrieval-augmented generation (RAG) has emerged as a leading approach for grounding language model outputs in external knowledge. Despite significant progress, current systems struggle with multi-hop reasoning over heterogeneous document collections. In this work we propose GraphRAG, a method that combines dense retrieval with graph-based evidence aggregation.

Our experiments on three benchmark datasets show that GraphRAG improves answer accuracy by 12.3% over dense retrieval baselines. The gains are most pronounced on questions requiring cross-document evidence synthesis.

2. Method

2.1 Architecture Overview

The model architecture consists of three components: a dense retriever, a graph constructor, and an answer generator. Each component can be trained independently or end-to-end. We found that end-to-end training provides marginal improvements at significantly higher computational cost.

Method	EM	F1	Recall
Dense Retrieval	45.2	52.1	48.7
BM25 + Rerank	41.8	48.9	45.3
GraphRAG (ours)	57.5	63.4	59.1

Table 1: Comparison of retrieval methods on the MultiHopQA benchmark.

3. Results

Our main results are summarized in Table 1. GraphRAG achieves the highest scores across all three metrics. The improvement is statistically significant with $p < 0.01$ under paired bootstrap resampling.

Figure 1: Accuracy improvement of GraphRAG over baselines across benchmark datasets.

4. Conclusion

We introduced GraphRAG, a method that augments dense retrieval with graph-based evidence aggregation. Our experiments demonstrate consistent improvements over strong baselines on multi-hop question answering benchmarks. Future work will explore scaling to larger document collections and incorporating structured knowledge sources.

References

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